# AGE 310: Instrumentation and Measurements

Dr. Kadeghe Fue, Department of Agricultural Engineering, SUA

Morogoro, Tanzania

**Abstract**

Lecture 7 addresses the Actuator stage, the final control element in an automated system. We analyze the conversion of electrical control signals into mechanical energy. The focus is on classifying actuators (Electrical, Hydraulic, Pneumatic), detailing the principles of solenoids and various electric motors (DC, Stepper, Servo), and highlighting the high-power density of hydraulic systems vital for heavy-duty agricultural machinery.

# Lecture 7: Actuators and Output Devices

## 7.1 Introduction to Actuators

### 7.1.1 Definition and Role

An **Actuator** is a device that converts an energy source (typically electrical, fluid, or pneumatic power) into physical motion or action (linear or rotary). It is the component that executes the command issued by the Controller (microcontroller, PLC, or ECU).

In the closed-loop control system, the actuator acts on the ”Plant” (the process being controlled).

### 7.1.2 Classification by Energy Source

1. **Electrical Actuators:** Convert electrical energy (voltage/current) into mechanical torque or linear force. They offer high precision and clean operation. (*Motors, Solenoids, Piezo Actuators*)
2. **Hydraulic Actuators:** Use pressurized, incompressible fluids (oil) to generate immense force and torque. They are indispensable for heavy work. (*Cylinders, Motors*)
3. **Pneumatic Actuators:** Use compressed air or gas. They offer fast actuation and clean operation but are limited in force output compared to hydraulics. (*Air Cylinders, Air Muscles*)

## 7.2 Electrical Actuators

### 7.2.1 Solenoids

A solenoid is the simplest linear electrical actuator, converting electrical energy directly into push/pull motion.

- **Principle:** An electromagnet (coil) is energized, generating a magnetic field that rapidly pulls a movable ferromagnetic core (plunger) into the coil.
- **Action:** Typically provides binary ON/OFF action.
- **Ag Application (Solenoid Valve):** Used extensively in irrigation systems to switch water flow ON or OFF to specific zones based on the microcontroller’s command. Also used as fuel shut-off valves in engines.

### 7.2.2 DC Motors

DC motors convert direct current into continuous rotation.

- **Brushed DC Motors:** Simple and cheap, but the brushes wear out over time, creating electrical noise (EMI) and requiring maintenance.
- **Brushless DC (BLDC) Motors:** Use electronic commutation (switches) instead of brushes. Highly efficient, high torque density, long life, and low EMI.
    - **Ag Use:** Ideal for powering UAV (drone) propellers, precision pumps, and modern conveyor belts.

### 7.2.3 Stepper Motors

- **Principle:** Rotates in discrete, fixed angular steps (e.g., $1.8^\circ$ per pulse).
- **Control:** Typically operated **open-loop** - the controller sends a command for $N$ steps, assuming the motor executes them accurately. No positional feedback is required (unless steps are missed).
- **Ag Use:** Essential for **Variable Rate Technology (VRT)**. Used to drive the metering rollers in precision planters and fertilizer spreaders, allowing precise, electronically controlled speed change.

### 7.2.4 Servo Motors

- **Principle:** A closed-loop system containing a DC motor, gearbox, and an integrated position sensor (Potentiometer or Encoder).
- **Control:** The controller sends a desired position ($\theta_{\text{SP}}$), and the internal circuit uses the sensor's feedback ($\theta_{\text{PV}}$) to correct its position instantly.
- **Ag Use:** Precise throttle control in generators, or small, high-precision robotic actions (e.g., controlling a camera gimbal on a drone, or adjusting the orientation of a robotic apple picker).

## 7.3 Fluid Power Actuators (Hydraulics and Pneumatics)

### 7.3.1 Hydraulics

Hydraulic systems use pressurized oil to transmit power. This offers the highest power density, critical for heavy agricultural machinery.

**Pascal's Law:** Force is directly proportional to the area of the piston and the fluid pressure.

$$F = P \times A$$

**Hydraulic Cylinders:** The most common actuator, providing linear motion (push/pull).
[Image of Hydraulic cylinder diagram showing piston, rod, and ports]

- **Single Acting:** Uses oil pressure to extend, relies on gravity or a spring for retraction. (*Example: Tipping trailer*).
- **Double Acting:** Uses oil pressure for both extension and retraction. (*Example: Tractor steering, front-end loader*).

***Calculation Example 1: Hydraulic Force***

A double-acting hydraulic cylinder used on a sugar cane harvester has a piston diameter ($D_p$) of $10$ cm and operates at a system pressure ($P$) of $20$ MPa ($20 \times 10^6$ Pa). Calculate the maximum extension force ($F_{\text{ext}}$) generated.

**Solution:**

- Piston Area ($A$):

$$A = \pi r^2 = \pi \left( \frac{0.10 \text{ m}}{2} \right)^2 \approx 0.00785 \text{ m}^2$$

- Maximum Extension Force ($F_{\text{ext}}$):

$$F_{\text{ext}} = P \times A = (20 \times 10^6 \text{ Pa}) \times 0.00785 \text{ m}^2$$

$$F_{\text{ext}} \approx 157,000 \text{ N} \approx 15.7 \text{ metric tonnes of force}$$

This massive force generation is why hydraulics dominate heavy agricultural equipment.

### 7.3.2 Pneumatics

Pneumatic systems use compressed air.

- **Pros:** Very fast, clean (no fluid leaks), high cycle rate.
- **Cons:** Air is compressible, leading to spongy or less precise positioning. Lower force output than hydraulics.
- **Ag Use:** Used in high-speed, light-duty applications such as milk pulsators in milking parlours, and vacuum/pressure systems for seed singulation in precision planters.

## 7.4 Motor Control Elements

### 7.4.1 Relays and Contactors

These are electrically actuated switches.

- **Relay:** Used for low-power switching, typically isolating a low-voltage microcontroller (e.g., $5$ V Arduino) from a medium-voltage load (e.g., $12$ V pump).
- **Contactor:** A heavy-duty relay designed to safely handle very high currents, used for starting large $3$-phase pump motors (common in large irrigation schemes in Tanzania).
- **Importance:** Provides electrical isolation and power amplification.

### 7.4.2 H-Bridge Motor Driver

To reverse the direction of a DC motor, the polarity of the voltage must be swapped. An H-Bridge circuit uses four switching elements (Transistors/MOSFETs) arranged in an 'H' configuration to allow the control circuit to easily switch the current path. [Image of H-Bridge circuit diagram showing four MOSFET switches, power supply, and DC motor]

### 7.4.3 Solenoid Valves (Final Control Element)

A solenoid valve is the essential control element in fluid systems.

- **Function:** Controls the flow of oil (Hydraulics), air (Pneumatics), or water (Irrigation).
- **Proportional Control:** Modern **Proportional Valves** do not simply switch ON/OFF. They use Pulse Width Modulation (PWM) control to position a spool, allowing continuous, variable flow control, which is essential for VRT fertilizer application.

***Calculation Example 2: Stepper Motor Position Control***

A Stepper motor used to drive a VRT metering auger has a step angle of $1.8^\circ$ (meaning $200$ steps per revolution). The auger requires $1500$ RPM for maximum fertilizer application rate.

1. Calculate the total number of pulses required for one full revolution of the motor.
2. Calculate the required frequency (pulse rate in Hz) from the controller to achieve $1500$ RPM.

**Solution:**

1. **Total Steps per Revolution ($N_{\text{steps}}$):**

$$N_{\text{steps}} = \frac{360^\circ}{1.8^\circ/\text{step}} = 200 \text{ pulses}$$

2. **Required Frequency ($f$):**

$$\text{Frequency} = \text{Steps per Revolution} \times \frac{\text{RPM}}{60 \text{ s/min}}$$

$$f = 200 \text{ steps/rev} \times \frac{1500 \text{ rev/min}}{60 \text{ s/min}} = 5000 \text{ steps/second} = 5 \text{ kHz}$$

The microcontroller must generate a $5$ kHz pulse train to the stepper motor driver to achieve this application rate.

## 7.5 Summary of Key Terms (Swahili/Context)

- **Actuator:** Kiwezeshaji (The device that executes the control command).
- **Solenoid:** Sumaku-umeme (Electromagnet used for linear ON/OFF motion).
- **Stepper Motor:** Motori ya hatua (Motor that moves in discrete, precise steps).
- **Servo Motor:** Motori ya mrejesho (Closed-loop motor with built-in feedback).
- **Hydraulics:** Mfumo wa maji mazito (System using pressurized oil for high force).
- **Contactor:** Swichi ya umeme mzito (Heavy-duty electrical switch/relay).
- **VRT:** Teknolojia ya Kiwango Tofauti (Variable Rate Technology - applying inputs based on prescription maps).

# Lecture 7: Review Questions

**SECTION A: Basic Concepts (5 Questions)**

1. Define an Actuator and explain its role in closing the loop of a typical control system.
2. Differentiate between the principles of operation of a *Stepper Motor* and a *Servo Motor* in terms of position control (open-loop vs. closed-loop).
3. State two distinct advantages that Hydraulic Actuators offer over Electrical Actuators in the context of heavy agricultural machinery (e.g., bulldozers, harvesters).
4. Explain the primary function of an H-Bridge circuit in controlling a DC motor.
5. Briefly describe the operation of a simple Solenoid valve used in a drip irrigation system (how electrical energy translates to physical action).

**SECTION B: Advanced Analysis and Calculations (10 Questions)**

6. **Hydraulic Force Calculation:** A double-acting hydraulic cylinder used to lift a tillage implement has a piston diameter of $15$ cm and a rod diameter of $5$ cm. If the system pressure is $18$ MPa, calculate:
   - The maximum extension force (Push).
   - The retraction force (Pull).
   - Explain why the retraction force is lower than the extension force.
7. **Stepper Motor Frequency:** A Stepper Motor with a $0.9^\circ$ step angle drives a pump. If the desired flow rate requires the motor to run at $120$ RPM, calculate the required pulse frequency in Hz.
8. **Relay Isolation:** An Arduino microcontroller provides $5$ V and a maximum of $20$ mA from a digital pin. Explain why a Relay or MOSFET driver is essential to control a standard $12$ V, $10$ A water pump. What is the technical term for this necessary process?
9. **Proportional Valve Control:** Explain how a *Proportional Solenoid Valve* uses Pulse Width Modulation (PWM) from a microcontroller to achieve variable flow control, rather than simple ON/OFF action. Give an agricultural application where this is vital.
10. **Pneumatics vs. Hydraulics:** A robotic fruit sorting machine requires a fast, low-force linear actuator to push fruit off a conveyor belt. Would you recommend a Pneumatic or a Hydraulic cylinder, and why? Explain the limitation that prevents the use of a Pneumatic cylinder for high-force applications.
11. **Servo Motor Feedback:** Explain the function of the position sensor (Potentiometer/Encoder) integrated into a Servo Motor. How does this sensor maintain the closed-loop nature of the servo?
12. **BLDC Motor Advantage:** Explain the key advantage of a Brushless DC (BLDC) motor over a traditional Brushed DC motor in the context of UAV (drone) operation, focusing on efficiency and electromagnetic interference (EMI).
13. **VRT Stepper Control:** A $1.8^\circ$ Stepper motor is used in a VRT fertilizer application system. If the tractor ground speed doubles, how must the controller adjust the pulse frequency to maintain the same fertilizer application rate per hectare?

14. **Contactor Selection:** An agricultural engineer must select a Contactor to start a $50$ kW pump motor operating at $400$ V ($3$-phase). Estimate the minimum rated current capacity required for the Contactor, assuming a motor efficiency of $90$% and power factor of $0.85$. (Formula for 3-phase power: $P = \sqrt{3} \cdot V \cdot I \cdot \text{PF}$).

15. **Hydraulic Safety:** In a single-acting hydraulic cylinder system used for tipping a trailer, explain why the operator must check that the relief valve is correctly set. What failure could occur if the relief valve pressure setting is too high?